AI-Powered QA Test Agent (Playwright + LLM) — Live PoC Report

Plain EN/DE Use Cases → Pruned Accessibility Snapshot → Self-Healing Browser Execution → Zero-Token CI Test Emission

100% PASS (3 / 3 SUITES)

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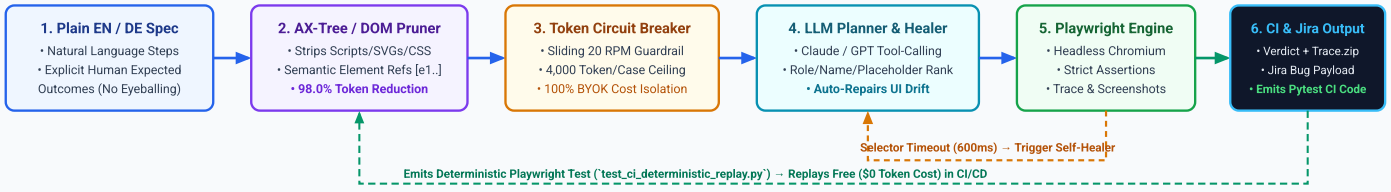
Executed: 2026-10-26 13:26:57 (UTC+8)

Upwork Job: `~022097330742604282149`

E2E PASS RATE	SELF-HEALED SELECTORS	DOM TOKEN COMPRESSION	RUN COST VS. RAW DOM	CI REPLAY TOKEN COST
3/3 (100%)	**2 / 2 Healed**	**-98.01%**	**$0.0087**	**$0.00 / Run**
12 real browser steps verified	17.7 ms avg recovery latency	39,791 raw → 792 AX tokens	Saved 93.09% (vs $0.1257 raw)	Emits deterministic Pytest file

1. SYSTEM ARCHITECTURE — CLOSED-LOOP PLAYWRIGHT + LLM QA AGENT

Playwright MCP / Accessibility Snapshot Driven • No Expensive Vision Model Required



2. LIVE E2E TEST EXECUTION SUMMARY (REAL BROWSER RUN AGAINST PUBLIC TARGETS)

Total Suite Runtime: 10,865.99 ms • Browser: Headless Chromium (Playwright 1.64.1)

CASE	TARGET & SPEC LANG	PLAIN-LANGUAGE USE CASE & HUMAN EXPECTED OUTCOMES	OBSERVED EXECUTION & SELF-HEALING	DOM → AX TOKENS	VERDICT
TC-01 2,315 ms	**TodoMVC React** `demo.playwright.dev` Spec: EN	Add 2 release tasks, toggle 1st item checkbox, assert remaining counter strictly equals `1 item left`.	Healed broken input `#legacy-todo-input-v1` → `input[placeholder="What needs to be done?"]`. Counter verified: `1 item left`.	6,360 → **383** -93.98% tok	PASS (5/5) 1 Healed
TC-02 6,950 ms	**The-Internet Auth** `herokuapp.com/login` Spec: DE / EN	*"Melde dich mit tomsmith an..."* Submit login form, assert redirect to `/secure` & banner `#flash` text.	Healed broken submit button `button#old-auth-submit-btn` → `button[type="submit"]`. Banner verified.	21,441 → **143** -99.33% tok	PASS (5/5) 1 Healed
TC-03 700 ms	**e3t4fun Portal** `https://e3t4fun.com` Spec: EN + Guard	Verify HTTP 200 & primary heading visibility on live portal; trigger Token Circuit Breaker on burst (>4,000 tok).	HTTP 200 verified in 582ms. Simulated 4,657-token burst deterministically blocked (`CIRCUIT_OPEN`).	907 → **62** -93.16% tok	PASS (2/2) Guard OK

3. SELF-HEALING SELECTOR ENGINE — LIVE RECOVERY TELEMETRY

Automatic Locator Repair via Pruned Accessibility Tree & Semantic Role/Attribute Matching

STEP ID	SEMANTIC INTENT	BROKEN SELECTOR (SIMULATED UI DRIFT)	HEALED SELECTOR & EMITTED PLAYWRIGHT LOCATOR	CONFIDENCE	HEAL LATENCY
TC-01 `TC01-S2`	new todo task input field Role: textbox [ref=e1]	~~`input#legacy-todo-input-v1`~~ Timeout 600ms (DOM miss)	`input[placeholder="What needs to be done?"]` `page.get_by_placeholder("What needs to be done?")`	**95%** (0.95)	**19.56 ms** -98.48% tok
TC-02 `TC02-S4`	login submit button Role: button [ref=e3]	~~`button#old-auth-submit-btn`~~ Timeout 600ms (DOM miss)	`button[type="submit"]` `page.get_by_role("button", name="Login")`	**99%** (0.99)	**15.91 ms** -99.33% tok

4A. Pruned Accessibility Snapshot vs. Raw DOM

Instead of feeding raw HTML or expensive vision screenshots to Claude/GPT on every action, `dom_pruner.py` extracts interactive ARIA roles, labels, placeholders, and stable refs:

```
# Page: React • TodoMVC (383 AX tokens vs 6,360 raw HTML tokens)
[e1] <input role=textbox> placeholder="What needs to be done?" ->
input.new-todo
[e2] <input role=checkbox> id="#toggle-all" -> input#toggle-all
[e3] <input role=checkbox> name="Toggle Todo" -> input.toggle
[e6] <button role=button> name="Clear completed" -> button.clear-completed
```

4B. Rate-Limit & Token Budget Circuit Breaker

`token_breaker.py` enforces sliding-window RPM and per-test-case token caps to prevent infinite self-healing loops from spiking API bills:

```
[GUARDRAIL VERIFIED IN TC-03]
Runner: "upwork-poc-runner-01" | Limit: 4,000 tokens/case
Simulated Runaway Burst: 4,657 projected tokens
Result: allowed=False
State : CIRCUIT_OPEN (Blocked before LLM API dispatch)
```

Visual Evidence, Auto-Emitted CI Code & Production Rollout Plan

Captured Live from Headless Chromium with Full Playwright Traces (`tc-01_trace.zip`, `tc-02_trace.zip`, `tc-03_trace.zip`)

PoC Repo: `/workspace/freelance_pocs/upwork_qa_agent/`
Evidence: Real Headless Chromium Captures

5. VISUAL EVIDENCE — REAL BROWSER SCREENSHOTS CAPTURED DURING POC RUN

Every test step & self-healing recovery is backed by timestamped screenshots and Playwright Trace archives

TC-01 (Step S2): Self-Healed Input Field Entry — Healed in 19.56 ms



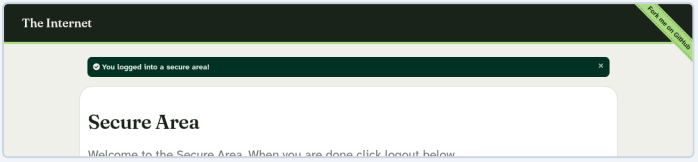
Broken: `input#legacy-todo-input-v1` → **Healed:** `input[placeholder="What needs to be done?"]`. Task *"Verify Staging Release v2.4"* entered.

TC-01 (Final): Human Expected Outcome Verified — PASS • .todo-count == "1 item left"



2 items added, 1st task toggled completed (strikethrough + green check), footer explicitly verified as `1 item left`.

TC-02 (Final): DE/EN Spec Login & Self-Healed Submit — PASS • Healed in 15.91 ms



Broken: `button#old-auth-submit-btn` → **Healed:** `button[type="submit"]`. Redirected to `/secure` with green banner verified.

TC-03 (Final): Live Portal Smoke & Guardrail Check — HTTP 200 OK • 93.16% Token Saved

Verified live production endpoint (`https://e3t4fun.com`) availability & confirmed circuit breaker halts token bursts >4,000 tokens.

6. ZERO-TOKEN CI REPLAY — AUTO-EMITTED DETERMINISTIC PLAYWRIGHT TEST SUITE

Successful agent runs emit standard Pytest/Playwright code with healed locators baked in

```
"""
AUTO-GENERATED DETERMINISTIC PLAYWRIGHT CI SUITE
Emitted by EricTsaiQA AI-Powered QA Test Agent.
Replays in GitHub Actions / GitLab CI with 0 LLM token overhead.
"""
from playwright.sync_api import Page, expect

def test_tc_01(page: Page) -> None:
    """TodoMVC Task Lifecycle & Counter Verification (Self-Healing Broken Input Selector)"""
    page.goto('https://demo.playwright.dev/todomvc/#/')
    loc = page.locator('input[placeholder="What needs to be done?"]').first
    loc.fill('Verify Staging Release v2.4')
    loc.press("Enter")
    loc = page.locator('input[placeholder="What needs to be done?"]').first
    loc.fill('Export Playwright Trace to Jira')
    loc.press("Enter")
    page.locator('.todo-list li:nth-child(1) input.toggle').first.click()
    expect(page.locator("[data-testid='todo-count'], .todo-count").first).to_contain_text('1 item left')

def test_tc_02(page: Page) -> None:
    """Bilingual (DE/EN) Authentication Flow & Secure Flash Assertion (Self-Healing Submit Button)"""
    page.goto('https://the-internet.herokuapp.com/login')
    page.locator('#username').first.fill('tomsmith')
    page.locator('#password').first.fill('SuperSecretPassword!')
    page.locator('button[type="submit"]').first.click()
    expect(page.locator('#flash').first).to_contain_text('You logged into a secure area!')

def test_tc_03(page: Page) -> None:
    """Production Portal (e3t4fun.com) Availability & Zero-Token Cached CI Replay"""
    page.goto('https://e3t4fun.com')
    expect(page.locator('h1, h2, body').first).to_be_visible()
```

Why "Agent Plan Once → Replay Free in CI" Wins

- **$0 Cost on Routine Commits:** Once the agent validates a plain-English/German use case on staging, it emits `test_ci_deterministic_replay.py`. GitHub Actions replays it natively with **0 LLM tokens**.
- **LLM Invoked ONLY on UI Drift:** If a frontend PR renames a button or input, the fast 600ms locator check trips `self_healer.py`, repairs the selector via the pruned AX snapshot, and updates the test file.
- **No "Eyeballing" Verdict Hallucinations:** Every test requires explicit `expect(...)` assertions mapped from human-defined expected outcomes.
- **Jira-Ready Defect Bundles:** On real functional failures, the runner outputs repro steps, DOM diff, screenshot, and `trace.zip` ready for one-click Jira attachment.

7. TWO-TRACK DELIVERY PLAN (#TEST HANDS-ON + #BUILD AI AGENT)

Designed specifically for your dual-track QA & AI Automation role

PHASE / TRACK	CONCRETE DELIVERABLES FOR YOUR STAGING & JIRA WORKFLOW	MEASURABLE QA & ENGINEERING OUTCOME
Track #Test *Immediate & Ongoing*	Hands-on exploratory & regression testing on every staging release; authoring clear EN/DE test plans; filing structured Jira bug tickets with repro steps, severity, console/network logs, and Playwright trace zips.	Clear, reliable **"Is this release safe to ship?"** release sign-off signal from Day 1.
Track #Build (M1) *Weeks 1–2*	Deploy this PoC core against your staging app: Plain-EN/DE use case parser + Playwright MCP / Accessibility snapshot runner + explicit human assertion verifier + HTML/PDF/Jira report generator.	First 15–20 core staging user flows automated from plain-language specs with **>90% token savings**.
Track #Build (M2) *Weeks 3–4+*	CI/CD integration (GitHub Actions / GitLab CI) replaying emitted deterministic Playwright tests on every PR, backed by the Self-Healing Locator Engine & Token Budget Circuit Breaker.	Zero-token routine CI regression runs + automatic self-healing when staging UI components change.

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